Revisiting *Remarks*: the learning of English nominals

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Generative linguistics has long been concerned with the descriptive adequacy and explanatory adequacy of theories. The typical approach is to propose a theory that first achieves descriptive adequacy, then, to varying degrees, discuss how it may relate to explanatory adequacy. The present paper illustrates a new approach: I propose a theory that achieves explanatory adequacy, by showing it can be learned by a child under a concrete theory of learning, then show that it also achieves descriptive adequacy. I do so by considering gerundive and derived nominals, as in Chomsky 1970.
[Note: This is work in active progress. So please excuse any errors, and feedback is always welcome!]

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1 Introduction

Since Chomsky 1965, the benchmarks for evaluating a theoretical proposal about grammar have been descriptive adequacy and explanatory adequacy. The descriptive adequacy of a theory is evaluated by the extent to which it makes the correct empirical predictions. The explanatory adequacy of a theory is evaluated by whether the proposed grammar can be learned by a child.

The standard logic of work in theoretical linguistics is to propose a theory that achieves descriptive adequacy. Then, one might show how the theory relates to explanatory adequacy, e.g. by showing that it conforms to, and bears on, Universal Grammar.[1]

The goal of this paper is to illustrate a new approach. The logic of this new approach is: propose a theory that achieves explanatory adequacy, by showing it is learnable by a child, then show that the theory also achieves descriptive adequacy. This approach is possible because of an emerging theory of learning, developed in Yang 2016 and subsequent work, which gives us a concrete way to evaluate learnability. There is already a small but growing body of work that takes this perspective of learning to understand various aspects of language (Kodner 2020, Legate & Yang 2025, Belth 2026, Lu et al. 2026, a.o.). The present paper is an attempt to integrate this perspective directly into theoretical syntax, by showing how we can propose theories that are learnable, while covering the same empirical grounds that syntacticians are interested in.

To illustrate this approach, I focus on a small corner of English syntax: nominalizations. Specifically, I take Chomsky's (1970) "Remarks on nominalization" as my starting point.

The central focus of Chomsky 1970 is the distinction between gerundive nominals, as in (1b), and derived nominals, as in (1c).[2]

(1) a. John refused the offer.
    b. John's refusing the offer
    c. John's refusal of the offer

Chomsky notes:

> Many differences have been noted between these two types of nominalization. The most striking differences have to do with the productivity of the process in question, the generality of the relation between the nominal and the associated proposition, and the internal structure of the nominal phrase.
>
> (Chomsky 1970: 15–16)

Gerundive nominals like (1b) can be formed freely from ordinary declarative clauses. Moreover, gerundive nominals display, in some way, the syntax and semantics of a vP. Gerundive nominals preserve the event semantics of a corresponding vP, including the thematic roles of the participants. For example, *John's collecting seashells* denotes a similar event as the clause *John collects seashells.* Gerundive nominals also contain structure typically associated with a vP, e.g. modification by adverbs and direct objects.

On the other hand, derived nominals such as those in (1c) are unproductive: many verbs, such as *read*

[*] I thank Charles Yang for extensive discussion on earlier versions of this manuscript. I am also indebted to Dan Brodkin, Alexander Hamo, Muhammed Ileri, Julie Legate, and the participants of Penn Syntax Lab for feedback on the accompanying talk.

[1] At its core, UG is really an interface between descriptively adequate theories and learnability by a child. Principles are added to UG insofar as they are deemed necessary for a grammar to be learnable. By committing to a theory of learning, though, we can ask in a more principled way what is and is not necessary to build into UG. More on this in the discussion.

[2] By "gerundive nominals", I always mean those of the form in (1b), consisting of a possessor and a phrase with *-ing*.

and *write*, do not have a corresponding derived form.[3] Furthermore, the meaning of a derived nominal may be idiosyncratically related to the meaning of a similar clause; compare the meanings of *John collects seashells* and the derived nominal *John's collection of seashells*. Derived nominals also have the internal structure of an NP: it is modified by adjectives rather than adverbs, and the verb-related head can only take PP complements, not direct objects like in gerundive nominals.

There has been a huge body of literature since Chomsky 1970 (Aronoff 1976, Abney 1987, Grimshaw 1990, Marantz 1997, Alexiadou 2001, Fu et al. 2001, Borer 2003, Alexiadou & Borer 2020, a.o.), which has greatly expanded the empirical picture around English nominalizations. In this paper, I will only be able to cover a subset of these empirical facts, but I hope it is nevertheless enough to illustrate how an explanatorily adequate theory may also achieve descriptive adequacy on par with existing theories.

The foundation of this approach is a theory of learning that allows us to concretely evaluate the learnability of a grammar. Thus, I will take care to introduce this theory of learning in section 2. Then, in section 3, I apply this theory of learning to examine gerundive nominals. I first provide a learning story for how gerundive nominals can be learned on the basis of the child's input. As my learning story produces a concrete analysis for the grammar responsible for generating gerundive nominals, I then show that this analysis makes desirable empirical predictions. That is, I first provide an account that achieves explanatory adequacy, then show that it is also descriptively adequate. In section 4, I do the same for derived nominals. Lastly, section 5 concludes, returning to some of the broader theoretical issues raised above.

2 The theory of learning

In this section, I introduce the theory of learning adopted here. I begin by introducing the notion of productivity in section 2.1, which becomes essential when we consider the learning of rules in section 2.2. In section 2.3, I show how we can understand the acquisition of a grammar as the learning of individual rules. In section 2.4, I discuss how this theory of learning leads us to a way to evaluate learnability. By the end of the section, we will have a practical method to evaluate the learnability of a syntactic proposal, which we will soon apply to the problem of gerundive and derived nominals.

2.1 The notion of productivity

To understand the present approach to learning, it is crucial to understand the notion of productivity.

First, some clarifications on terminology. I use "rule" to mean anything a child learns. Although rules are most commonly discussed in the context of morphological or phonological rules, I use them as a general term for learned linguistic knowledge, which can include lexical items or syntactic categories. Moreover, the existence of a rule is independent from it being productive. Productivity simply has to do with what the domain of the rule is, i.e. either the rule applies to a memorized set of items, or to an open class (and thus may generalize to unseen items). There are both productive and unproductive rules. These aspects will be made clear in the ensuing discussion.

We will start with a familiar example of a productive rule. English productively forms the past tense of verbs with the suffix *-ed*. We can state this rule as in (2).

(2) **Rule**: English regular past tense

V + [PAST] → V-*ed* (productive over V)

[3] Chomsky also mentions what he calls "mixed nominals" like *John's writing of the book*, which have the structure of derived nominal, but with the *-ing* suffix on the verb. I provide an analysis of mixed nominals in Appendix A.

This rule is productive over the domain of verbs. This productivity has behavioral consequences: when an English speaker encounters a new word, they form its past tense using *-ed*. This is true for adult speakers (who were presumably responsible for *Google-Googled*), as well as children (Berko 1958). This behavior is notable because it indicates that, somewhere in the speaker's knowledge, they must have a rule like (2). Once they learn a new lexical item belonging to the category V, the rule automatically applies to it, even if that item was not part of the evidence that originally motivated the rule. This capacity to go beyond one's input is what productive rules buy us.

In contrast, some rules are unproductive. For example, several English verbs have a past tense in *-ought*. We can write the relevant rule as in (3). This rule is not productive over any domain; instead, it is memorized for individual verbs.

(3) **Rule**: English irregular past tense in *-ought*

{seek, think, bring} + [PAST] → verb with rime in /ɔt/ (memorized for each verb)

In other words, we can understand productivity as capturing whether a rule operates on an open domain such as V, or a finite domain such as a set of specific lexical items.

This idea of productivity extends beyond the familiar sort of morphological/phonological rules. For example, we can think of selection between syntactic objects as being productive or unproductive. Transitive verbs like *enjoy* productively select the category DP. This rule is productive over the open domain of DP. In contrast, *rely* selects a PP headed by the specific lexical item *on*. The domain of this rule is thus a single lexical item. We can state these selectional rules as in (4).[4]

(4) **Rules**

a. [enjoy DP] (productive over D)
b. [rely on] (memorized)

Selection is one of the few domains of syntactic knowledge that we can naturally think of productivity as being salient. In fact, part of why syntacticians rarely think about productivity is perhaps because the vast majority of what we typically count as syntax deals with productive rules, so we take productivity for granted. This is, in a large part, the result of architectural design. Anything that is unproductive, which is typically stated over lexical items, is pushed to the lexicon, so the syntax remains clean and productive. This is one of the primary motivations behind Chomsky 1970 in the first place. Under the grammatical architecture of the time, the grammar is divided into the lexicon and the transformational component. Nominalization was an interesting problem precisely because, while both derived and gerundive nominals bear some relationship with a declarative clause, the former is unproductive while the latter is productive. Thus, Chomsky proposed to relegate the relationship between derived nominals and the clause into the lexicon, while the relationship between the clause and gerundive nominals is one of productive transformation.[5] This distinction in their productivity was really the core motivation of Chomsky 1970, though subsequent work has never really fully developed this idea.

So, why does all this matter for learning?

Because productivity and unproductivity are two natural outcomes of learning. In particular, productive rules are those rules about the input that can be generalized, while unproductive rules are those that cannot be generalized.

[4] As one might notice, these selectional rules have two components, the selector and the selectee. In principle, rules can be productive or unproductive over either component. The rule in (4a) is productive over the selectee but (trivially) unproductive over the selector (since we are stating it as a rule governing a single lexical item), while the rule in (4b) is unproductive over both components.

[5] See Chomsky 2020 for a more recent discussion of this view.

When the child encounters patterns in their input, they start by memorizing them, i.e. store them as unproductive pieces of knowledge. For example, upon encountering a few instances of *thought* used in a past tense context to mean *think*, the child can simply memorize that the past tense of *think* is *thought*. The same goes for *run-ran*, but also *dance-danced* and *laugh-laughed* at first. These unproductive rules do not extend beyond the child's memorized experience.

But for some of these patterns, the child generalizes them as a productive rule. For example, all English speakers have only heard a finite number of verbs occur with *-ed* in our lifetime, yet we know that *-ed* applies to all verbs, including those we have not encountered. Clearly this knowledge cannot have been purely memorized from experience. Instead, we must have generalized some rule that operates over all verbs, including those we have not heard. So when exactly do we do this generalization?

2.2 How a rule is learned

We can now, finally, consider how a rule is learned.

In the case of unproductive rules, they are simply memorized. How much evidence is required for the memorization of a single rule is a matter of general cognitive factors, e.g. working memory capacity, and consequently subject to individual variation. But this can be done on the basis of relatively few instances in the input.

The more interesting case is the learning of productive rules. When do children generalize a productive rule from unproductive, memorized examples? The Tolerance Principle (TP; Yang 2016) answers precisely this question. The TP, stated in (5), provides a criterion for whether there is sufficient evidence for a rule to be learned as productive over some domain.[6]

(5) The Tolerance Principle

Let a rule R be defined over some domain D of size N. Let M be the number of examples[7] attested in the input that follow R.

Define a threshold

$$\gamma_N = N - \frac{N}{\ln N}.$$

Then, R is productive over D if and only if

$$M \geq \gamma_N.$$

The TP gives a concrete mechanism for when children generalize memorized examples to productive rules. Suppose a child has hypothesized some rule R over the domain of verbs. Say they have $N = 100$ verbs in their vocabulary. If the number of these verbs that are attested in the input following rule R is at least γ_N, then rule R is learned as productive over verbs. Otherwise, R is not productive, and the child simply memorizes each of the attested examples individually.

So learning consists of this iterated process of first memorizing examples in the input, hypothesizing general rules over those examples, then verifying whether those rules can be learned as productive according to the Tolerance Principle.[8]

[6] Some terminological notes. The Tolerance Principle was originally introduced by Yang (2016: 64, (11)) as a criterion to determine whether there are too many exceptions in a paradigm to tolerate a productive rule. What I call the "Tolerance Principle" here was presented then as the "Sufficiency Principle" (Yang 2016: 177, (3)), which is a corollary to the original formulation of the Tolerance Principle and the version more useful to learning. These terms are often used interchangeably in the literature. I have also defined the threshold in the Sufficiency Principle ($N - \theta_N$ where $\theta_N = \frac{N}{\ln N}$) as γ_N, since we are now counting the number of positive examples, rather than exceptions.

[7] This counts the number of types, not tokens.

[8] See Yue 2026 for a model of word learning that captures this process of rote memorization followed by a search for productive rules.

2.3 How a grammar is learned

So far, we have considered the learning of a single rule. However, there is a natural extension of this approach from the learning of a single rule to the learning of a grammar, which can be understood as an interdependent set of rules.

The important insight here is that the acquisition of a grammar takes place over the course of time. This dimension of time is crucial. Unlike as implicitly assumed in most theoretical models of language acquisition (Chomsky 1981, Gibson & Wexler 1994, Perfors et al. 2011), the child is never faced with the daunting task of picking out the one correct grammar in its entirety, given all the input they receive. Instead, the child builds their grammar incrementally: at each point in time, the child learns the rules that can be stated in terms of their existing knowledge and are compatible with their input; then, the learning of these rules form the basis of new rules; and so on. The adult grammar is the end result of this learning process.[9]

This incremental process is mechanistic. At each point, the child learns the rules that occur most frequently in their input that they can state in terms of their existing knowledge. As a direct consequence, the child learns more "basic" rules, which are most robustly attested in their input, early on, before learning progressively more complex rules, which require basic rules to formulate. This view of learning thus allows us to construct a discovery procedure for language (Chomsky 1957: 50–51); see Yang et al. 2026.

2.4 How to evaluate learnability

The view of learning described above directly informs how, as linguists, we can evaluate learnability. Our starting point is to look at the child's input. How we evaluate the learnability of a proposed rule depends on whether or not it is productive. If the proposed rule is unproductive, then as long as the child receives some attestations of the rule in the input, it is learnable.

On the other hand, if the proposed rule is productive, then we count the number of positive attestations (M) in the input and compare that number to the threshold γ_N stated by the Tolerance Principle in (5). If M is greater than or equal to γ_N, then the rule is learnable as a productive rule. Otherwise, the rule is not productive, and the individual attestations must then be memorized (or, one must search for productivity within a subset of the domain).

To evaluate the learnability of a (part of a) grammar, one must identify the individual learned rules that make up that part of the grammar, and evaluate the learnability of each of those rules. I will demonstrate this approach later in evaluating the learnability of the part of the grammar that generates gerundive and derived nominals.

Before moving on, some practical notes. I use a 6.8-million token corpus of English child-directed speech (MacWhinney 2000) as a proxy for the input of a child.[10] Furthermore, since children learn the core rules of their language by a young age (around age 3), they do so on a limited vocabulary size; the specific vocabulary size varies among children, but around 800 words is reasonable for a typical child (Hart & Risley 1995). Verbs comprise on the order of 15% of these early words (Bates et al. 1994, Frank et al. 2021), so we take the child to know 100 verbs. Since children learn the frequent words earlier (Goodman et al. 2008), I use the most frequent 100 verbs in my dataset as a proxy for the child's vocabulary of verbs. This method of using a frequency-trimmed corpus as a proxy for the child's vocabulary is standard in this line of work (e.g. Björnsdóttir 2021, Belth 2023, Kodner 2023, Thoms et al. 2025, Belth 2026, Lu et al. 2026).

[9] Another way of putting this is that the learning of a grammar is the result of local steps, rather than some global decision made from aggregating all of the child's received input. This is the more psychologically plausible way of thinking about learning (see Stevens et al. 2017 for more discussion in the context of word learning).

[10] For reference, children hear around 5 million words per year, subject to individual variation.

3 Learning gerundive nominals

We will devote most of our energy to understanding how gerundive nominals, rather than derived nominals, are learned. This is for two primary reasons.

First, as noted by Chomsky (1970: 16–17), gerundive nominals, but not derived nominals, are productive.[11] In our view of learning, it is the productivity of rules, rather their lack thereof, that need to be explained. We thus need to show that the productivity of gerundive nominals are learnable from the input.

Second, the learning of gerundive nominals is particularly interesting, as it constitutes a poverty of the stimulus problem. Although they have been heavily featured in linguistics research, gerundive nominals occur very rarely in a child's input. More specifically, while gerundive nominals consisting of a possessor and a verb in *-ing* are relatively frequent, gerundive nominals that include additional verbal structure such as direct objects are exceedingly rare. In my search, I identified 220 instances of gerundive nominals in the input. Of those, 66 are formed from one of the most frequent 100 verbs, i.e. a verb in the child's vocabulary. This token count corresponds to 26 unique verbs (i.e. types). Of these 66 gerundive nominals, only 9 of them occur with a direct object, corresponding to 7 unique verbs.

In other words, a child may only encounter around 20 examples of gerundive nominals containing a verb in their vocabulary and additional verbal structure (e.g. direct objects) in the course of their early development. Nevertheless, on the basis of these few examples, they are able to reliably learn a grammar that generates gerundive nominals productively.

Our learning story must then provide a solution to this problem, i.e. show how gerundive nominals can be learned despite their rarity in the input. I will show that this problem can be resolved by capitalizing on the insight that learning is incremental. The child learns rules incrementally, and their grammar at any point is the composition of these individual rules. Gerundive nominals are generated by a part of this grammar, i.e. the relevant subset of rules a child has learned. The task of the linguist is then one of "reverse engineering": we want to decompose a speaker's knowledge of gerundive nominals into the individual rules that generate them, with each of those rules being individually learnable. I will do precisely this.

My account can be summarized as follows. Gerundive nominals are learned as the composition of two rules: the category ingP, and the selection of ingP by the possessive head.[12] The child learns the first rule, the category ingP, on the basis of a large amount of evidence from the input: all examples of *-ing* with verbal structure, not just gerundive nominals. This rule can be learned as a productive over vP. Then, the child learns the second rule, that possessive heads selects ingP, on the basis of much fewer examples: the specific instances of gerundive nominals. This rule is only productive over the small class of possessive heads, thus requires less evidence from the input. In other words, the first rule is harder to learn but has more evidence, while the second rule has less evidence but is easier to learn.[13] This decomposition resolves the poverty of the stimulus problem and explains how the child can learn gerundive nominals.

In the rest of this section, I will illustrate this proposal in more detail, then show that the grammar that results from learning also achieves the desired empirical coverage.

[11] The subsequent discussion will make clear exactly what is productive and over what domain.

[12] Plus the child's prior knowledge, e.g. basic vocabulary and basic categories like NP, DP, VP, vP. This knowledge may be innate or previously learned.

[13] We can also understand why the child learns these two rules in this order. Children learn more frequent rules earlier than less frequent rules; this is the same logic as learning more frequent words earlier than less frequent ones. The first rule is attested much more frequently: children hear phrases with *-ing* much more than a particular structure in which that phrase occurs (the latter being a proper subset of the former). Thus, naturally, the child learns the more general rule of the category ingP before learning any particular selection rule over ingP.

3.1 The category ingP

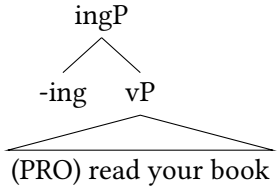
Ultimately, we are interested in understanding how gerundive nominals are acquired. However, to do so, we must zoom out to consider the broader empirical picture that constitutes a child's input. This is because, while the linguist's point of departure is to understand "gerundive nominals", the child has no such prior concept. Instead, they are simply trying to make sense of their input. In order to reason about acquisition, we must take the perspective of the child.

If we look at the child's input, what we immediately see is an abundance of examples involving *-ing*. Consider (6), a sample from child input taken from my dataset. Gerundive nominals like (6f) are only a tiny subset of instances of *-ing*. My search identified 105,498 instances of tokens ending in the *-ing* suffix.[14] Of these, 98% were labeled as participles, most of which are progressives like (6a), while only 2% were gerunds of any sort, including (6c-6f). Of these 105k instances of *-ing*, 80k occurred with a verb in the top 100, which covers 99 unique verbs.[15]

(6) a. you're [going to hurt me] (progressive)
b. he came towards me [waving] and [saying "spoon"] (secondary predication)
c. I enjoyed [reading your book] (complement to *enjoy*)
d. thank you for [being our friend] (complement to *for*)
e. you were thinking about [being a teacher] (complement to *about*)
f. let's tell them about your [breaking your finger] (gerundive nominal)

What immediately jumps out from these examples is that there is a unit that consistently recurs across varying environments. Internally, this unit always involves *-ing*, alongside material associated with a vP (i.e. a verbal head and additional material like direct objects).[16] Externally, this unit is reused in different syntactic environments, with a distribution distinct from a vP. This combination of a consistent internal structure and recurrence across different syntactic environments motivates a child to posit an abstract grammatical object, of which each of these units are instantiations. This is exactly what a syntactic category is.[17]

Each of the units in (6) consistently involve the lexical item *-ing*. *-ing* can thus be understood as the head of the category, which I call "ingP". In each of these units, the head *-ing* selects a vP containing some verb. The structure of the ingP in (6c) is given in (7).[18]

(7)


Note that just encountering the examples in (6) and observing the ingPs is not sufficient fo learning a productive rule over verbs. On the basis of (6), the child may simply memorize lexical rules: *-ing*

[14] Code for this and all other counts reported in this paper is available here.

[15] The only verb in the top 100 not attested with *-ing* is *excuse*.

[16] Throughout the paper, I take the relevant phrases in *-ing* to involve vP, rather than VP. This is because the unit with *-ing* is interpreted as involving a subject, and as we will see later, this subject is sometimes overt. Since subjects are standardly assumed to be introduced by a higher functional head like v (Kratzer 1996), I adopt this vP analysis. This choice is based on the presence of subjects alone, but I have not considered in detail the implications this additional layer and head may have for learning.

[17] More work is needed in general to precisely state what it means to learn a syntactic category, which I do not resolve here. However, the idea that a syntactic category is fundamentally characterized by its internal structure and external distribution is straightforward.

[18] I remain neutral on the treatment of the subject in the various uses of ingPs. In gerundive nominals, specifically, it may be base-generated within the vP then raise to specD, or there may be a PRO in the vP controlled by a DP in specD.

selects {*go, wave, say, read, be, break*}.

But if we consider the larger input, the child has sufficient evidence to learn a productive rule over verbs.

Recall the Tolerance Principle, stated in (5). For a rule stated over a domain of size N, the child needs to encounter at least $\gamma_N = N - \frac{N}{\ln N}$ attestations of that rule to learn it as productive over the domain. We want to evaluate whether *-ing* productively forms ingPs over all vPs. Of the most frequent $N = 100$ verbs, $M = 99$ of them are attested with *-ing* in the input. This far surpasses the threshold for productivity, which is $\gamma_{100} = 79$ for $N = 100$. Thus *-ing* can be learned as productive over verbs.

Furthermore, the child can learn that the rule is stated over the phrasal category (vP), not just V. To do so, they may use direct objects as a diagnostic for verbal structure. I thus count how many transitive verbs occur in *-ing* with a direct object.[19] Out of the 83 transitive verbs in the top 100 verbs, $M = 80$ of them occur in *-ing* with a direct object, which easily surpasses the threshold of $\gamma_{79} = 65$.[20]

Thus, the child can learn that *-ing* productively selects vPs. We can state this more formally as the rule in (8).

(8) **Rule:** the category ingP

$[_{\text{ingP}}$ -ing vP] (productive over vP)

This analysis captures three core properties of ingPs: 1) their internal verbal structure follows from the fact that they contain a vP; 2) their distinctive external distribution follows from the fact that they are a distinct category (i.e., *-ing* rather than v/V projects); and 3) they can be productively formed from any vP.[21]

3.2 D_{poss} selects ingP

Now, the child has learned ingP as a syntactic category. What does this buy us? Say the child encounters a phrase involving *-ing* in some syntactic environment. Now, they can identify that phrase as instantiating an ingP. Moreover, if they have not previously encountered ingPs occurring in this syntactic environment, the child can store a new rule stating that ingPs can occur in this environment. Contrast this with the case if the child had not learned ingPs. Every time the child encounters some phrase involving *-ing* in some environment, they need to relearn the internal structure of that phrase from scratch, as there is no object such as ingP to which they can make reference. In other words, learning the category of ingP buys the child the ability to make reference to ingPs in learning further rules. In particular, as selection rules are stated in terms of a selector and the head of the selectee, having the category ingP allows the child to simply learn a rule about the head *-ing* directly, bypassing the question of productivity over verbs.

[19] In principle, a child can make use of other diagnostics for vP structure, not just the presence of a direct object. The reason I chose to count direct objects is practical. Since the CHILDES corpus is not parsed, I cannot simply search for vP structure. Instead, all searching must be done over linear sequences of words or parts-of-speech tags. I thus restricted my search to nominals (pronouns or determiner + noun combinations, with optional modifiers) following the verb in *-ing*, then manually checked the output for true gerundive nominals.

[20] This diagnostic based on direct objects only applies to the domain of transitive verbs, which is why here I evaluate the TP only on transitive verbs. For intransitives, either the child makes use of other verbal material (e.g. adverbs), or the evidence with transitives is sufficient for them to infer the rule applies to all verbs, since they know intransitives independently never occur with a direct object.

[21] Although ingPs are an unfamiliar syntactic category, none of its properties are independently unique. *-ing* realizes a functional head as a suffix on the verb, much like the past tense suffix *-ed*. The morphological productivity of *-ing* is really a consequence of its being able to productively select vPs, much like the T head *-ed*. Though, despite this similarity, ingPs cannot be TPs, as they occur in distinct distributions (aside from a very small amount of overlap with nonfinite TPs headed by *to*). One of the perhaps more notable properties of ingPs is that it has only one lexical head, *-ing*. However, already in English, we propose functional categories, like T and C, that are instantiated by very few lexical items; in fact, Neg may be a standardly assumed category that only has one lexical item, *not*.

Once we understand this logic, it becomes surprisingly simple to understand how gerundive nominals are learned. The child simply needs to learn that ingP occurs with possessive structures. Or, in other words, possessive determiners select ingP.

There are seven possessive determiners in standard English: *'s, my, our, your, his, her, their*. These form the subclass of determiners D_{poss}. Of the $N = 7$ elements in D_{poss}, all $M = 7$ occur in the input with an ingP.[22] Thus the child can easily learn the rule in (9). Note that this rule is not productive over the larger class of D.

(9) **Rule:** D_{poss} selects ingP

$[_{DP}\ D_{poss}\ \ ingP]$ (productive over D_{poss})

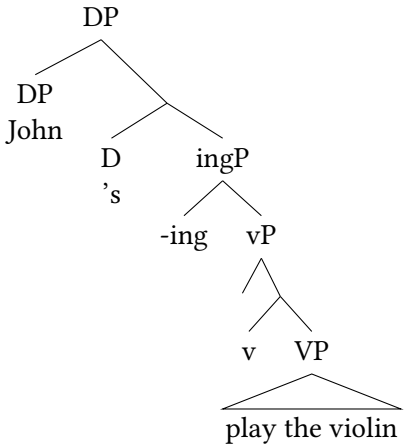
Together, the two rules in (8) and (9) are sufficient to generate gerundive nominals. (10) shows how gerundive nominals can be generated.

(10) a. $[_{vP}$ play the violin] (prior knowledge of vP)
b. $[_{ingP}$ -ing $[_{vP}$ play the violin]] (rule in (8): the category ingP)
c. $[_{D'}$'s $[_{ingP}$ playing the violin]] (rule in (9): D_{poss} selects ingP)
d. $[_{DP}$ John's playing the violin] (prior knowledge of possessives)

Thus, the only rule specific to gerundive nominals that the child needs is (9), which is learned over a small class of elements. Their pre-existing knowledge and their broader knowledge of ingPs from rule (8) take care of the rest. I have thus shown how gerundive nominals can be learned as the composition of two rules, each of which are individually learnable.

While I have taken as my starting point to provide a plausible learning story for gerundive nominals, this story about *how* gerundive nominals are acquired arrived at a particular analysis of *what* is acquired. The structure of a gerundive nominal, under my analysis, is given in (11).

(11)



In section 3.3, I will explore the consequences of this analysis, ultimately showing that this analysis of gerundive nominals not only achieves explanatory adequacy by being learnable, but also makes the desired empirical predictions, thus is also descriptively adequate. But before doing so, I will take a quick detour to mention the various other uses of ingPs, beyond gerundive nominals.

[22] I am counting within the 220 instances of gerundive nominals in the input, rather than the 9 instances of gerundive nominals additionally with a direct object. This is because the learning that ingPs can contain verbal structure is handled by the previous rule in (8), so children do not need to relearn that ingPs can contain verbal structure when learning each individual rule governing how ingPs may be used. This again illustrates how the division of labor between the two rules allows gerundive nominals to be learnable.

3.2.1 Selection of ingP by other heads

We have just seen that gerundive nominals can be understood as resulting from the rule in (9), that D_{poss} selects ingP. The other uses of ingPs, e.g. those in (6), can also be understood as resulting from various similarly stated selectional rules about ingP.

Consider (12), which represent various structures in which ingP occurs. These examples are not from my input data, but include the structures underlying the examples in (6).

(12) a. John [$_{VP}$ is [$_{ingP}$ playing the violin]].
b. John [$_{VP}$ enjoys [$_{ingP}$ playing the violin]].
c. John makes money [$_{PP}$ by [$_{ingP}$ playing the violin]].
d. John [$_{VP}$ [$_{VP}$ walked in the room] [$_{ingP}$ playing the violin]].
e. The [$_{NP}$ [$_{NP}$ student] [$_{ingP}$ playing the violin]] was happy.

To account for the data in (12), the child simply needs to learn the various rules around how ingPs can be used.

In each of (12a–c), the ingP occurs as a complement to some head. The relevant rules to generate each of these structures are simple: some head selects an ingP. These rules only need to be memorized for the specific heads that the child encounters as taking an ingP in their input; we can write them as in (13).[23] [Note: Counts will be added at a later date.]

(13) **Rules**

a.	[$_{VP}$ be ingP]	(memorized)
b.	[$_{VP}$ enjoy ingP]	(memorized)
c.	[$_{PP}$ by ingP]	(memorized)

The child now has enough knowledge to generate (12a–c). To illustrate this, consider the derivation of (12b), given in (14). Only two new rules making reference to ingPs, (8) and (13b), needed to be learned.

(14) Derivation of (12b)

a.	[$_{vP}$ play the violin]	(prior knowledge of vP)
b.	[$_{ingP}$ playing the violin]	(rule in (8): the category ingP)
c.	[$_{VP}$ enjoy [$_{ingP}$ playing the violin]]	(rule in (13b))
d.	John enjoys playing the violin.	(prior knowledge of TP)

A similar logic can apply to learning the rules that generate (12d–e). Just like how they learn heads that select an ingP, the child learns the phrasal categories that may combine with an ingP (i.e., in an adjunction relation). The rules can be stated as in (15). Since these rules are stated over phrasal categories, they are necessarily productive over those categories and must be learned as such. I do not further discuss how these rules are learned here.[24]

[23] It is possible that, just like in the case of gerundive nominals, there exist productive rules over certain domains that can subsume some of these lexically learned rules. For example, Ps may be able to productively select ingPs, as 20 out of the most frequent 20 Ps in the child's input are attested with an ingP complement. The only Ps that do not occur before ingP are those that are exclusively locative. In any case, the existence of further productive rules over these ingP-selecting heads is not important for present purposes.

[24] Ordinarily, selection is understood as a relationship between a selector and selectee, where the selector is a head. On the other hand, adjunction is a relationship between two phrases. However, rules about the selection and adjunction possibilities in a language are all more general instances of rules about what items can be combined (i.e. Merged). There is thus a natural way to view both these sorts of rules as similar learning problems.

(15) **Rules**

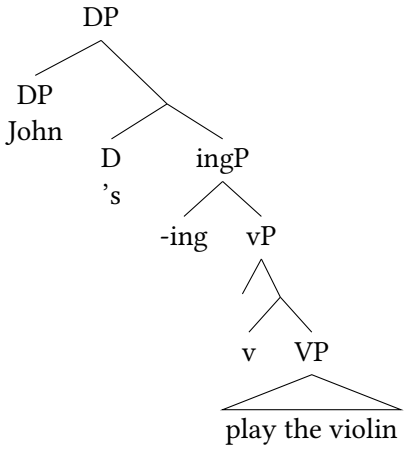
a. $[_{VP}$ VP ingP]
b. $[_{NP}$ NP ingP]

3.3 Empirical consequences

[Note: Much of this section will be reorganized and expanded upon in future drafts.]

Now we return to gerundive nominals. I will explore the various empirical predictions made by the analysis of gerundive nominals as in (11), reprinted in (16).

(16)



Many of the empirical facts discussed here are familiar from previous treatments. My goal is not to introduce new data, but rather to show that my analysis, which is independently motivated by learning, reproduces existing empirical coverage. Moreover, in the ways where I differ from existing analyses, I show that my analysis makes desirable empirical predictions.

First, my analysis in (16) resembles standard accounts that take gerundive nominals to contain VP/vP (e.g. Abney 1987, Kratzer 1996, Marantz 1997). As a result, my analysis covers the same empirical ground in the relevant areas; I thus do not focus on the empirical consequences of this particular aspect of the analysis. Instead, I will focus on the features of my analysis that diverge from many existing analyses.

3.3.1 The source of nominality

A notable feature of the present analysis is that the possessive D directly selects an ingP. As a consequence, in no point in the derivation is ingP an NP.

This feature of my analysis provides an easy explanation of one often observed fact about gerundive nominals: they never occur in other DP structures aside from the possessive, as in (17a). This is in contrast to derived nominals, as in (17b).

(17) a. *The playing the violin (by John)
b. The playing of the violin (by John)

Under my analysis, this is because the rule that generates gerundive nominals is stated as a rule productive over possessive subclass of D. It is not productive over the larger class of Ds. Thus it does not generate ingP as complement to any other D heads.

Moreover, we can easily understand why gerundive nominals do not display nominal internal structure, e.g. resist adjectival modification. APs and other nominal modifiers attach to NP: since there is no NP in the structure of gerundive nominals, we do not expect these nominal modifiers to occur.

3.3.2 The shared internal syntax and semantics of ingPs

As we have seen, ingPs are used in a variety of environments beyond gerundive nominals. Some have tried to account for a subset of these uses (e.g. Pires 2007). However, no analysis, to my knowledge, extends to the full range of uses of ingPs considered here.

My analysis provides a way to precisely characterize the unity among these various uses of *-ing* phrases: they are all instantiations of the same category ingP. From a learning perspective, this is a natural approach, as children do not divide the various occurrences of *-ing* phrases into *a priori* labels such as "gerund" or "participle". Instead, they treat them as instances of one grammatical object, then learn the distribution of that object.

A consequence of this analysis is that the properties of ingPs that follow from their internal syntax and semantics automatically apply across these various uses. We will examine the internal syntax and semantics of ingPs here.

First, an ingP contains a vP.[25] This accounts for the collection of properties of ingPs associated with verbal structure, including adverbial modification (18a), presence of auxiliaries (18b), passivization (18c), raising structures (18d). It can be easily verified that ingPs across different environments share these syntactic properties.

(18) a. John's deliberately ignoring the warning
b. John's having read the book
c. John's having been examined by his doctor
d. John's seeming confused

Likewise, if the various instances of *-ing* phrases all instantiate the category ingP, their basic internal semantics should remain constant, with interpretive differences arising from their external construction.[26]

This is indeed the case. In particular, the ingPs uniformly introduce an event predicate with an external argument. Interpretive differences arise when ingPs enter into various external structures and its subject argument is saturated in different ways.

Consider the meaning of an ingP. An ingP can simply inherit the denotation of the vP. While *-ing* determines syntactic category, it makes no semantic contribution. So the denotation of the ingP *playing the violin* is something like (19).

(19) $[\![\textit{playing the violin}]\!] = \lambda x.\lambda e[\text{play}(e, \text{the violin}) \wedge \text{agent}(e, x)]$

This predicate then enters semantic composition with varying external material and has its subject saturated in different ways. We will briefly consider some of these structures.

3.3.3 The various external constructions with ingPs

We will consider the following constructions in which ingPs occur: gerundive nominal, complement to control verb, subject position, adjunction to DP, and adjunction to VP. My focus is not on defending any particular analysis for these structures, but just to put forth one plausible analysis and show that

[25] I use vP here to stand in for a relatively large verbal projection that contains the subject, as well as the functional structure needed for aspectual modification and passivization.

[26] By "basic internal semantics", I roughly mean the event denotation and how it enters into semantic composition.

the mechanism of interpretation is not specific to ingPs but can be shared by other categories that occur in the same positions.

From a learning perspective, the point here is that the child learns the various selectional rules involving ingPs, which generate the various syntactic structures. The semantics then follows, based on general rules of semantic composition.

First, we consider gerundive nominals, which involve ingPs in a possessive structure. Although gerundive nominals are syntactically exceptional among possessive structures in that it involves an ingP, rather than NP, complement to D, the semantics of the possessive structure remains fairly similar. The general semantic function of the possessive D is to saturate some relation associated with its complement with the individual in its specifier (Vikner & Jensen 2002, Partee & Borschev 1998). With a nonrelational noun like *book*, possessive D introduces an underspecified relation between John and a book. With a relational noun like *father*, possessive D simply saturates the argument of that relation with the individual in its specifier. For example, the possessive construction *John's father* can be interpreted as in (20).[27]

(20) a. $[\![\textit{father}]\!] = \lambda y.\lambda x[\text{father}(x, y)]$
b. $[\![\textit{John's father}]\!] = \lambda x[\text{father}(x, \text{john})]$

The gerundive nominal may be interpreted similarly to a relational noun. The ingP supplies a predicate with an open individual argument, which is then saturated by the individual introduced by the specifier of the possessive head. The internal semantics of the ingP then interprets this individual as the subject of an event. A derivation is given in (21).

(21) a. $[\![\textit{playing the violin}]\!] = \lambda x.\lambda e[\text{play}(e, \text{the violin}) \wedge \text{agent}(e, x)]$
b. $[\![\textit{John's playing the violin}]\!] = \lambda e[\text{play}(e, \text{the violin}) \wedge \text{agent}(e, \text{John})]$

This unity in the semantic function of the possessive D between gerundive nominals and ordinary nominal possessive structures was already noted by Abney (1987: 206). In both cases, the possessive D saturates a relation between the individual in its specifier and some argument of the predicate in its complement. The only difference with the gerundive nominal is that this relation is between an individual and event, rather than between two individuals.

Nevertheless, as we have seen, the meaning of the gerundive nominals follows directly from the composition of the ingP with a possessive structure. This means that a child need not learn a gerundive-specific meaning for possessive D, in order to interpret gerundive nominals.[28]

We can apply a similar logic to understand the other constructions involving ingP. I will go through some of these constructions.

As (22a) shows, ingP may occur as the complement to control verbs. This position may be occupied also by nonfinite TPs, as shown by (22b), with a similar interpretation. The interpretive mechanism is thus a property of control structures, rather than unique to ingPs. (23) provides a sample derivation of how (22a) may be interpreted, based on the treatment of control by Chierchia (1989).[29]

[27] I focus on the predicative part of the possessive structure, setting aside further operations in interpreting the full DP.

[28] This depends a little on one's semantic analysis for possessive D. If one wants to maintain that possessive D operates on relations between individuals, then for the case of gerundive nomianls, the child may need to learn that it can also operate on relations between an individual and an event. However, I am inclined to think that, once the child learns gerundive nominals as a grammatical structure, their knowledge of its meaning easily follows.

[29] For the complement of a control verb and the subject position, I have assumed an analysis where ingP contains PRO. One may adopt similar analyses for other structures, e.g. gerundive nominals and DP-modification, where ingP contains a PRO, which is then lambda-abstracted over to result in an open predicate (**landau2013ch3**, **chierchia1990**). Again, I am not committed to the particulars of any of these analyses.

(22) a. John likes [$_{\text{ingP}}$ PRO playing the violin]
b. John likes [$_{\text{TP}}$ PRO to play the violin]

(23) a. $[\![\textit{like}]\!] = \lambda P.\lambda x.\text{like}(x, P)$
b. $[\![\textit{PRO playing the violin}]\!] = \lambda x[\text{play}(e, \text{the violin}) \wedge \text{agent}(e, x)]$
c. $[\![\textit{John likes PRO playing the violin}]\!] = \lambda x[\text{like}(\text{John}, [\![\textit{PRO playing the violin}]\!])]$

A similar treatment applies to ingPs in the subject position, as in (24a). Here, PRO also occupies the subject position inside the ingP, but receives a generic interpretation instead.

(24) a. [$_{\text{ingP}}$ PRO playing the violin] is John's lifelong dream.
b. [$_{\text{TP}}$ PRO to play the violin] is John's lifelong dream.

Next, ingPs also occur in adjunction structures. (25a) shows that ingPs can adjoin to DP as a post-nominal modifier, on par with other predicative phrases as in (25b–c). Semantically, these structures may be interpreted as involving the modifier composing with the DP via predicate modification. A derivation for the relevant parts of (25a) is given in (26). Before step (26c), existential closure over the event argument of the ingP converts it to an ordinary property of individuals, allowing predicate modification.

(25) a. The [$_{\text{NP}}$ [$_{\text{NP}}$ student] [$_{\text{ingP}}$ playing violin]] delighted John.
b. The [$_{\text{NP}}$ [$_{\text{NP}}$ student] [$_{\text{AP}}$ passionate about violin]] delighted John.
c. The [$_{\text{NP}}$ [$_{\text{NP}}$ student] [$_{\text{CP}}$ who was playing violin]] delighted John.

(26) a. $[\![\textit{student}]\!] = \lambda x[\text{student}(x)]$
b. $[\![\textit{playing violin}]\!] = \lambda x.\lambda e[\text{play}(e, \text{violin}) \wedge \text{agent}(e, x)]$
c. $[\![\textit{student playing violin}]\!] = \lambda x[\text{student}(x) \wedge \exists e[\text{play}(e, \text{violin}) \wedge \text{agent}(e, x)]]$

A similar analysis applies to ingPs adjoined to vP, as in (27a). Syntactically, this is an adjunction structure on par with adjunction to DP. Semantically, this involves predicate modification over the event argument, rather than the individual argument (i.e., "secondary predication"). Relevant parts of the derivation are given in (28).

(27) a. John [$_{\text{vP}}$ [$_{\text{vP}}$ left the room] [$_{\text{ingP}}$ cursing his teacher]]
b. John [$_{\text{vP}}$ [$_{\text{vP}}$ left the room] [$_{\text{AP}}$ angry at his teacher]]

(28) a. $[\![\textit{cursing his teacher}]\!] = \lambda x.\lambda e[\text{curse}(e, \text{his teacher}) \wedge \text{agent}(e, x)]$
b. $[\![\textit{left the room}]\!] = \lambda x.\lambda e[\text{leave}(e, \text{the room}) \wedge \text{agent}(e, x)]$
c. $[\![\textit{John left the room cursing his teacher}]\!]$
$= \lambda e[\text{curse}(e, \text{his teacher}) \wedge \text{agent}(e, \text{John}) \wedge \text{leave}(e, \text{the room})]$

I do not discuss further constructions in which ingP can appear, but they can be analyzed on par with the constructions discussed above. In particular, the constructions involve adjunction of the ingP can be treated similarly to (25) and (27), while constructions involving complement to lexical heads can be treated similarly to (22).[30] We can understand each of them as slotting ingPs into the relevant syntactic

[30]One of the more interesting constructions is the complement to the verb *be*, specifically the progressive interpretation. Although typically the progressive construction is treated as instantiating aspectual syntax, I am inclined to treat the progressive on par with other constructions where *be* takes a predicative complement (e.g. an AP in *I am angry at him*), then locate the progressive interpretation elsewhere. In particular, when ingP occurs as complement to *be*, its subject may be identified as the matrix subject of *be*, perhaps due to raising. This particular configuration is associated with an aspectual interpretation. On the other hand, when the matrix subject in a *be*-ingP structure is different from the subject in ingP, the resulting interpretation is different (e.g. the equative/specificational *My job is playing violin*). One natural question is then whether the two uses of *be*-ingP involve the same syntax, specifically with respect to how the ingP subject is interpreted.

structure and letting compositional semantics take care of interpreting it.

3.3.4 ingPs with overt subjects

Another consequence of my analysis of is that it provides a natural way to understand the *-ing* phrase in expressions such as (29). These phrases are typically called "acc-ing gerunds", following Horn (1975). I will thus adopt this as a neutral term to refer to these kinds of phrases.[31]

(29) [$_{\text{ingP}}$ Him reading the book] delighted me.

I propose that the *-ing* phrases as in (29) are ingPs with an overt subject. Up until now, we have only seen ingPs without an overt subject, which I call "subjectless ingPs". However, since ingPs make available a subject position, we expect that the subject may be overt, which is indeed the case. This is in line with previous work that treats acc-ing gerunds as the overt-subject verions of what I call subjectless ingPs (Horn 1975, Reuland 1983, Pires 2007).

If my proposal is correct, then acc-ing gerunds and subjectless ingPs are of the same category. We thus expect them to appear in the same distribution, with any differences in distribution independently explainable. I will show that this is indeed the case: the differences in distribution between acc-ing gerunds and subjectless ingPs can be independently explained by their difference in subject realization.

First, see that acc-ing gerunds and subjectless ingPs overlap in many of the core environments in which ingP occurs. (30) shows this for complements to control verbs and prepositions, adjuncts to DP, and subject position.

(30) a. John enjoys [$_{\text{ingP}}$ (Bill) playing the violin].
b. John left the room without [$_{\text{ingP}}$ (Bill) seeing the teacher].
c. [$_{\text{ingP}}$ (Bill) playing the violin] delights John.

There are several environments, however, in which only the subjectless ingP but not the acc-ing gerund may occur. This includes adjunction to DP, as in (31a), and adjunction to vP, as in (31b).

(31) a. The [$_{\text{NP}}$ [$_{\text{NP}}$ student] [$_{\text{ingP}}$ (*Bill) playing violin]] delighted John.
b. John [$_{\text{VP}}$ [$_{\text{VP}}$ left the room] [$_{\text{ingP}}$ (*Bill) cursing his teacher]]

These contrasts can be explained by the presence of an overt subject in acc-ing gerunds but not subjectless ingPs. Both of these constructions involve predication modification, specifically DP modification and secondary predication. By nature, NP modification involves an adjunct whose open predicate is identified with the open predicate of the phrase to which it is adjoined. Therefore, an ingP that has already saturated that argument with an overt subject cannot enter into these modification constructions.

The analogous behavior of other categories participating in these constructions show that this restriction on the subject is not unique to ingPs. APs are another category that may occur in both constructions. (32) shows that APs also may not occur with a subject in the two adjunction structures.

(32) a. The [$_{\text{NP}}$ [$_{\text{NP}}$ student] [$_{\text{AP}}$ (*Bill) angry at his teacher]] annoyed John.
b. John [$_{\text{vP}}$ [$_{\text{vP}}$ left the room] [$_{\text{AP}}$ (*Bill) angry at his teacher]].

A full treatment of *be*-ingP here is beyond the scope of my paper, but see Déchaine 1995 for an analysis of *be* in roughly a similar vein.

[31] For present purposes, I leave aside the question of why the subject receives accusative case. This is a question beyond ingPs, as the subject of other small clauses uniformly receive accusative case in the same positions.

The reverse pattern occurs in some environments, e.g. the complement to verbs such as *see*. Here, the acc-ing gerund is possible, while the subjectless ingP is not.

(33) John saw [$_{\text{ingP}}$ *(Bill) playing the violin]

This is likewise a property of the environment, rather than ingPs. This can be illustrated again by considering the parallel behavior of APs in this environment, as in (34). This restriction on the subject is likely because the verb *see* requires a complete event to be perceived. The unsaturated predicate is incompatible with the semantics of *see*.

(34) John saw [$_{\text{AP}}$ *(Bill) angry at the teacher].

My analysis expects that a similar reasoning can be applied to other cases where acc-ing gerunds and subjectless ingPs diverge in distribution. This is true to my knowledge, though I do not further treat every such case here.

From here onwards, I will treat acc-ing gerunds as ingPs with an overt subject, or "overt-subject ingPs". This treatment of acc-ing gerunds will allow us to understand some other facts about ingPs.

3.3.5 ingPs are not nominal

In section 3.3.1, I discussed one feature of my analysis: the source of nominality in gerundive nominals come from a possessive D head taking ingP as its complement, then projecting a DP. This allowed us to understand the lack of NP structure within the gerundive nominal and the incompatibility with other Ds.

Another consequence of this feature is that ingPs are not syntactically nominal, i.e. neither an NP nor DP. Since I have argued that acc-ing gerunds are in fact ingPs with an overt subject, it follows that acc-ing gerunds are not nominal. This is in contrast to analyses that treat acc-ing gerunds as DPs (e.g. Abney 1987), but captures a widely shared intuition that acc-ing gerunds are more "clause-like" than gerundive nominals (Abney 1987, Reuland 1983, Johnson 1988, Pires 2007). This section will deal with some other observations about the distribution of acc-ing gerunds, which will simultaneously provide additional support for the treatment of acc-ing gerunds as ingPs, as well as the treatment of ingPs as distinct from DPs in general.

First, note that ingPs cannot be analyzed as TPs, cf. some analyses that try to account for the clause-like properties of acc-ing gerunds by treating them on par with *to*-infinitives (**cite**). This is because their distributions are simply too different. Aside from a few verbs that select both an ingP and *to*-infinitive (e.g. *start, continue, try, like, remember*), heads that select an ingP do not generally also allow a *to*-infinitive. Some of these contrasts are shown in (35). These differences in distribution between ingPs and nonfinite TPs are really about the selectional properties of the individual heads and cannot be reduced to any semantic generalization (just consider the synonymous verbs with different selectional behavior).

(35) a. John enjoys [$_{\text{ingP}}$ playing violin] / *[$_{\text{TP}}$ to play violin].
b. John keeps [$_{\text{ingP}}$ cursing his teacher] / *[$_{\text{TP}}$ to curse his teacher].
c. John avoids [$_{\text{ingP}}$ playing violin] / *[$_{\text{TP}}$ to play violin].
d. John recalls [$_{\text{ingP}}$ playing violin] / *[$_{\text{TP}}$ to play violin].
e. John attempts *[$_{\text{ingP}}$ playing violin] / [$_{\text{TP}}$ to play violin].

I thus do not further consider the possibility of treating ingPs as TPs.

The more promising comparison is between ingPs and DPs. This is because ingPs and DPs highly overlap in their distribution, so much so that both overt-subject ingPs (acc-ing) and gerundive nominals (poss-ing; Horn 1975) are traditionally classified as "gerunds".

For example, ingPs occur in many of the positions typically associated with nominals, including the subject position and the complement to various DP-taking verbs and prepositions, as in (36) and (37).

(36) a. [$_{DP}$ His leaving] surprised me.
b. [$_{ingP}$ Him leaving] surprised me.

(37) a. I anticipated [$_{DP}$ his leaving].
b. I anticipated [$_{ingP}$ him leaving].

However, as a reviewer of an earlier version of my analysis pointed out to me, the occurrence of ingPs in these "nominal" positions is not enough evidence for a child to learn ingPs as nominals, since these positions also permit other categories, as in (38) and (39).

(38) a. [$_{CP}$ That he left] surprised me.
b. [$_{TP}$ For him to leave] surprised me.

(39) a. I anticipated [$_{CP}$ that he would leave].
b. ?? I anticipated [$_{TP}$ him to leave].

The same problem holds for linguists: the occurrence ingPs in these positions should also not be enough evidence to analyze ingPs as nominal.

In fact, we can tease apart the distribution of ingPs and DPs by considering environments that permit one but not the other. One such environment is adjunction to CP, or the absolute construction. (40) shows that while ingPs, both overt-subject and subjectless, may occur in this position, DPs may not. This contrast serves as a clean case where the distributions of ingPs and DPs come apart.

(40) a. [$_{ingP}$ Having left the store], John ate a cookie.
b. [$_{ingP}$ The clerk having left the store], Mary stole a cookie.
c. *[$_{DP}$ The clerk's having left the store], Mary stole a cookie.
d. *[$_{TP}$ (For) the clerk to leave the store], Mary stole a cookie.

Another syntactic environment that teases apart the distribution of ingPs and DPs is the complement of certain verbs. While many lexical heads select both ingP and DP, there are also many lexical heads that select DP, but do not permit ingPs. (41)–(43) provide some examples.[32]

(41) a. *I expected [$_{ingP}$ him leaving early].
b. I expected [$_{DP}$ his leaving early].
c. I expected [$_{TP}$ him to leave early].

(42) a. *The scandal necessitated [$_{ingP}$ him leaving the company].
b. The scandal necessitated [$_{DP}$ his leaving the company].
c. *The scandal necessitated [$_{TP}$ him to leave the company].

[32]It is not an accident that these are relatively rare verbs, and the judgments may vary between speakers. Since these verbs occur less frequently in the input, any lexical rule stated for the verb may be learned more inconsistently across speakers. Thus these are the cases that would display the most variation in which complements they select on a lexical basis. The reason the expressions in (b) are consistently acceptable across speakers is because productivity over DP is the most easily learned, and gerundive nominals are simply a DP.

(43) a. *I recommended [$_{\text{ingP}}$ him leaving immediately].
b. I recommended [$_{\text{DP}}$ his leaving immediately].
c. ?? I recommended [$_{\text{TP}}$ him to leave immediately].

These are, in my opinion, the constructions that most straightforwardly establish how acc-ing gerunds, as an ingP, and gerundive nominals, as a DP, belong to different distributional categories. In addition, there have been ahost of other properties that have been used to show that acc-ing gerunds are more "clause-like" than gerundive nominals. These are discussed in Abney 1987: 171–181. These properties include plural agreement and restrictions on extraction, which I will discuss briefly.

First, conjoining two gerundive nominals triggers plural agreement on the verb, as shown in (44a). However, conjoining two overt-subject ingPs does not trigger plural agreement, as in (44b). In this respect, ingPs pattern like CPs, as in (44c). This could be explained if agreement is property of DPs but not other categories.

(44) a. [$_{\text{DP}}$ John's coming] and [$_{\text{DP}}$ Mary's leaving] *bothers/bother me.
b. [$_{\text{ingP}}$ John coming] and [$_{\text{ingP}}$ Mary leaving] bothers/*bother me.
c. [$_{\text{CP}}$ That John came] and [$_{\text{CP}}$ that Mary left] bothers/*bother me.

(Abney 1987: 175, (182))

Another pattern involves extraction. Gerundive nominals seem to be harder to extract out of than overt-subject ingPs, as indicated in the contrast in (45). This is consistent with general restrictions against extraction out of complex DPs. The fact that ingPs do not display such a restriction again suggests that they are not DPs.

(45) a. *The city that we remember [$_{\text{DP}}$ his describing t].
b. The city that we remember [$_{\text{ingP}}$ him describing t].

(Abney 1987: 178, (188))

Both of these phenomena can be understood as resulting from further rules that operate on DPs, but not ingPs. However, a satisfactory treatment of these facts would require a concrete analysis of what those rules are and how they are learned, which is beyond the scope of the present paper. Thus I will not discuss them further.

I have thus shown that ingPs and DPs indeed constitute two different categories, and acc-ing gerunds are ingPs with an overt subject. This treatment supports the broader analysis developed in this section: ingP supplies a stable internal verbal structure, including a position for a subject, which may or may not be overt. Other aspects of the syntax and semantics of expressions involving ingPs are due to the external construction in which the ingP occurs.

3.4 Interim summary: gerundive nominals

In this section, I have provided an account of gerundive nominals, which achieves both explanatory adequacy (i.e., is learnable by a child) and descriptive adequacy (i.e., makes the correct empirical predictions).

Furthermore, in setting up the problem of learning gerundive nominals, we also identified it as an instance of a poverty of the stimulus problem. Gerundive nominals occur rarely in a child's input, yet all English speakers robustly learn them as part of their grammar. I showed that children can successfully learn gerundive nominals as the composition of two rules: the category ingP and the selection of ingP by possessive determiners. The first rule is harder, in that it involves productivity over a large domain, but children have abundant evidence for it. The second rule has limited evidence, due to the rarity of gerundive nominals, but is easier to learn, as it involves productivity over a much smaller domain.

Thus an apparent poverty of the stimulus problem is resolved by an appropriate decomposition of the learning problem.

This account of *how* gerundive nominals are learned results in a concrete analysis of *what* is learned. My analysis converges in several aspects with existing theoretical accounts of gerundive nominals (e.g. with respect to the presence of vP), thus covers the same empirical ground. In the ways that my analysis differs (e.g. having ingP as its own category), it captures several empirical facts better than existing accounts. Moreover, my account unifies various uses of *-ing*, beyond gerundive nominals, in the shared category of ingP.

4 Learning derived nominals

In the previous section, we took the perspective of child language acquisition to examine gerundive nominals, asking what rules that can be learned on the basis of the input data that can generate gerundive nominals.

This section takes a similar approach to examining derived nominals. The question here is what rules a child may acquire in order to produce utterances such as (46).

(46) $[_{\text{DP}}$ The destruction of the city by the army]

I argue that surprisingly little construction-specific knowledge needs to be learned in order to produce derived nominals. The child only needs to learn the individual derived nouns via ordinary word learning. Specifically, they need to learn the category of a word like *destruction* as a noun and its meaning as a predicate of events. This, along with prior knowledge, is sufficient to account for the basic syntax and semantics of derived nominals.

Before presenting the positive learning case for derived nominals, I begin by comparing the learning problem of derived nominals with that of gerundive nominals. This is makes explicit the ways in which gerundive nominals are productive, while derived nominals are not.

4.1 What is not learned

The acquisition of gerundive nominals began with the acquisition of the syntactic category ingP. There are two important properties of ingPs: they are productive over verbs, and they contain vP structure. The child received sufficient input to support both properties.

Neither of these two properties can be learned for derived nominals. First, no individual nominalizing suffix is productive over verbs. The most frequent nominalizing suffixes, aside from *-ing* and agentive *-er*, are *-tion*, *-ment*, and *-al*. These suffixes each only occur with one verb in the most frequent 100: *fixation*, *movement*, and *trial*. This is, of course, far below the threshold of $\gamma_{100} = 79$ required for productivity over verbs.

Second, these derived nouns do not occur with material indicative of vP structure. Derived nouns never occur with direct objects or adverbs; they must instead introduce arguments through PPs and take nominal modifiers. Thus the child has no evidence for a rule that establishes a structural relationship between derived nominals and vP.[33]

Taken together, the input evidence for derived nominals supports no rule that is productive over vP, unlike for gerundive nominals. So, what rules do the child have evidence for, and how do these learned rules allow us to understand the properties of derived nominals?

[33]Both of the components discussed here are important for the difference between derived and gerundive nominals. This is especially clear when we consider mixed nominals: the *-ing* suffix is productive over V, yet the rule is stated over the word rather than phrase, thus the structure of mixed nominals patterns like derived rather than gerundive nominals. Mixed nominals are discussed in Appendix A.

4.2 What is learned

Now, I will discuss what is learned regarding derived nominals: the individual derived nouns. The various syntactic and semantic properties follow.

4.2.1 Derived nouns as category N

I start with the syntax. For a derived nominal such as (46), the only thing that needs to be learned to understand its syntax is the category of the derived noun *destruction*.[34] Since *destruction* appears in the structural distribution of an N (e.g. following determiners, takes nominal modifiers), the child can easily learn that it is an N. This is all just ordinary word learning.

The syntactic properties of derived nominals related to its nominal structure follow. *Destruction* projects an NP, thus is modified by APs and PPs, and can be selected by various Ds.[35]

4.2.2 The eventive meaning of derived nouns

On the semantics side, the child learns the meaning of *destruction*. In particular, the child learns that *destruction* denotes a predicate of destroying events.

This is again an instance of word learning. Just as the child can learn what events the verb *destroy* describes from how it is used, the child can likewise learn what events *destruction* describes from its use.

We can represent the meaning of the noun in (47), where the predicate destroy describes destroying events.

(47) $[\![\textit{destruction}]\!] = \lambda e[\text{destroy}(e)]$

We can thus understand the child's relevant knowledge of *destruction* in three parts. First is knowledge of its category. Since *destruction* is an N, it enters into nominal syntactic structures, with the corresponding compositional semantics of those structures. Second is the meaning of the lexical item *destruction* as a predicate of destroying events.

Third is the child's conceptual knowledge of what constitutes a destroying event. They know that destroying events typically consist of some change undergone by an affected entity, often with some cause, and may take place over time. This is world knowledge about destroying events, not tied to a particular lexical item. The lexical meaning of *destruction* as a predicate over these types of events is the interface between the linguistic knowledge of a lexical item and the general knowledge that may be accessed by this item.

This division of labor of a child's knowledge of *destruction* also provides a natural way to understand the similarities between the verb *destroy* and noun *destruction*. Even if the two lexical items are learned separately, they are both learned to predicate over the same type of events, i.e. destroying events.

4.3 Consequences

4.3.1 From derived nouns to derived nominals

Once the syntactic category and lexical meaning of the derived noun has been learned, no special learning is required to constructthe larger derived nominals. Because *destruction* is an N, it enters into the syntactic structures available to nouns. Because *destruction* denotes a predicate of events, the modifiers of the noun are interpreted as modifiers of those events.

[34] I use "derived nominal" to refer to the larger construction as in (46), while "derived noun" refers specifically to the lexical head such as *destruction*.

[35] There are some other restrictions on derived nominals, e.g. their inability to be pluralized and occur with certain demonstratives, but many of these properties are due to the semantics of derived nouns as denoting events rather than individuals.

First, since *destruction* is an N, it enters into nominal structures. For example, it may take PP complements, and once it projects an NP, it may host PP modifiers and be selected by D. This generates derived nominal structures such as (48).

(48) $[_{\text{DP}}$ The destruction of the city by the army in June]

Semantically, since the noun makes an event variable available, PPs may introduce relations of that event and an individual. *Of* introduces an underspecified relation, which is resolved with a patient-like interpretation, as that is the most salient participant in a destroying event. *By* introduces an agent-like relation, due to the lexical meaning of *by* (i.e. same in passives). Since the NP predicates over events, it is compatible with a temporal modifier such as *in June*. The denotation of the NP in (48) is thus (49).

(49) $[\![$*destruction of the city by the army in June*$]\!]$
$= \lambda e[\text{destroy}(e) \wedge \text{R}(e, \text{the city}) \wedge \text{by}(e, \text{the army}) \wedge \text{time}(e, \text{June})]$

Since derived nouns may predicate over eventualities, they also permit event-oriented modification. For example, they permit agent-oriented modifiers as in (50a) or aspectual modifiers as in (50b).

(50) a. The deliberate destruction of the city
b. The destruction of the city in three hours

These interpretations do not require the modifiers to combine with syntactically verbal structure, and instead follow from the lexical meaning of *destruction* as a predicate of events, rather than individuals.[36]

4.3.2 The morphological relationship between *destroy* and *destruction*

One of the features of the present analysis is that it does not make use of the morphological relationship between *destroy* and *destruction*. *Destruction* is learned as an independent lexical item, which enters into nominal syntax and has event semantics. This is a somewhat significant departure from much recent work on derived nominals (Alexiadou 2001, Borer 2003, 2014, Wood 2023, a.o.), which takes the morphological relationship between *destroy* and *destruction* as its starting point.[37]

To be clear, adopting a learning approach does not preclude an analysis that recognizes a morphological relationship between *destroy* and *destruction*. This relationship can be easily learned by a child. The learning process may proceed as follows. The child initially learns *destruction* as a whole word. Once they encounter enough instances of *-ion* in different words, they are able to segment it as a suffix, and

[36] The properties illustrated here belong to the larger cluster that Grimshaw (1990: ch. 3) associates with complex event nominals (CENs). Grimshaw represents the participants in CENs as grammatical arguments, rather than as modifiers of the kind assumed here. Nothing in the present account depends on this choice: a child may simply learn the argument structure of a noun as part of its lexical representation. However, note that one important motivation for Grimshaw's analysis is the purported obligatoriness of participants under CEN readings. This claim, as well as the categorical status of the broader cluster of CEN properties, has been challenged (Reuland 2011, Grimm & McNally 2013, Moulton 2014). I therefore treat the participants in derived nominals as modifiers that are resolved through contextual or conceptual knowledge.

[37] If we abstract away from the particulars of analyses, however, the approach here is quite compatible with some of the broader intuitions behind DM approaches to nominalization. Marantz (1997: 15) states: "nominalizations like 'destruction' and 'growth' are never 'verbs' at any stage in the derivation, and thus DPs like those in [examples of derived nominals] are not transformationally related to sentences...". Both DM and the present approach maintain this idea. For me, *destroy* and *destruction* are individually learned but share a similar meaning. DM goes a step further to encode this similarity into a shared grammatical object, i.e. the root. More broadly, a core motivation of DM, as in Marantz 1997, was to challenge the view that the word was the boundary between a productive syntax and unproductive lexicon. This is maintained in our approach: the lexicon can be understood as those unproductive rules that are learned individually for a lexical item, while the generative component consists of the productive rules that operate on abstractions over those items; those rules may apply to units below or above the word.

subsequently isolate the root *destruct*. This is just lexical learning at the morpheme level. Once *destruction* can be decomposed, the child can identify a relationship between *destruct* and the verb *destroy*, on the basis of their similarity in form and meaning. See Ileri & Hamo 2026 for an implementation of such a learning model under the framework of Distributed Morphology.

Although such a relationship is learnable, it is not necessary to derive the properties of derived nominals.

We have already seen that the basic syntax and semantics of derived nominals can be understood as a consequence of learning individual derived nouns, specifically their syntactic category and semantic denotation. Identifying the morphological connection with *destroy* may be helpful for the child to learn the meaning of *destruction*, but children can learn the meaning of words even without such a connection.

This raises another question: if event meaning can be learned independently of morphological relatedness, why do the two appear so closely related?

One of the most discussed properties of derived nominals is their meaning, that they can be interpreted as denoting an event with complex internal structure (the "CEN" interpretation, following Grimshaw 1990). In English, it seems to hold that all nouns that allow a CEN reading also happen to be morphologically related to a verb, whether by means of an overt nominalizing suffix (e.g. *destroy-destruction*) or by sharing the same form (e.g. *murder*). One natural way of capturing this generalization is to say, roughly, that the CEN interpretation of nouns arises only when the noun structurally contains a verb.

On the other hand, if, as I propose, the CEN meaning of a derived noun can be learned independently, without making use of its morphological relationship with the verb, I seem to be missing this generalization. Moreover, we may expect to find CENs that are not morphologically related to a verb.

I argue that this is not a bug, but rather a feature of my analysis. While I do not provide a full treatment here, I will point out some directions. The core observation is that the strong generalization above is really a fact about English (and perhaps other languages, though I do not know enough to comment on them). For example, Huyghe et al. (2017) discuss a class of nouns in French that are not morphologically related to verbs, yet display the properties associated with CEN readings. If the French facts are indeed correct, an account that requires morphological relatedness for CEN meaning cannot provide a general explanation of CEN interpretations.

Moreover, we can understand why the strong generalization seems to hold in English without resorting to building it into the architecture of grammar. First, English allows the formation of nouns from verbs through the productive *-ing* and numerous other unproductive nominalizing suffixes. This explains the vast majority of CEN nouns that are related to a verb. Furthermore, and perhaps more crucially, English permits productive zero conversion, as illustrated by Clark & Clark's (1979) classic examples *to porch the newspaper* and *to Houdini one's way out*. Thus, although I have not confirmed this idea, it provides one diachronic explanation: there may be lexical items that truly began as underived CEN nouns, but subsequently gained their verbal use due to productive zero conversion in English. This would erode precisely the class of CENs that are not also morphologically related to a verb. Further candidates for this historical class may be verbs that were back-formed from nouns, e.g. *donation-donate* and *diagnosis-diagnose*.

These processes thus provide possible historical explanations of the English correlation, without requiring that the correlation itself be built into the architecture of grammar.

See Wood 2023: sec. 5.1.5 for a particularly clear exposition of these issues.[38]

[38]The view in Wood 2023 and related work is that the morphological relationship between the verb and the derived noun is necessary to understand the CEN properties of the derived noun. This is because Wood is working under a system that assumes a strong connection between the meaning and category of a word. In particular, nouns cannot introduce the sort

4.4 Interim summary: comparing gerundive and derived nominals

I have thus provided an account for how gerundive and derived nominals are learned. The learning of gerundive nominals consists of two steps: the child first learns the category ingP, which is productive over V/vP; they then learn a second, smaller step, that possessive determiners select ingP. These two rules, each of which I have shown to be learnable, together generate gerundive nominals. I have then discussed how the analysis of gerundive nominals that result from these two rules make desirable empirical predictions.

In contrast, derived nominals follow a very different learning path. The input around derived nominals does not support any productive rule over V/vP, thus the child simply cannot learn any rule that forms derived nominals from vP. Instead, the child learns individual lexical items. Specifically, they learn that the individual derived nouns have the syntactic category of N and the semantic denotation as a predicate of events. These two pieces of lexical knowledge derive the basic syntax and semantics of derived nominals.

We have thus provided an account for both gerundive and derived nominals that, by being learnable, achieves explanatory adequacy; and by making the right empirical predictions, achieves descriptive adequacy.

5 Discussion

Since Chomsky 1957, our model of the language system looked something like Figure 1. On the basis of their input data, the child constructs a grammar, which generates the linguistic behavior that we observe. The child constructs this grammar via some learning mechanism (the language acquisition device, or LAD), which operates on some existing linguistic knowledge (Universal Grammar, or UG) and the input they receive.

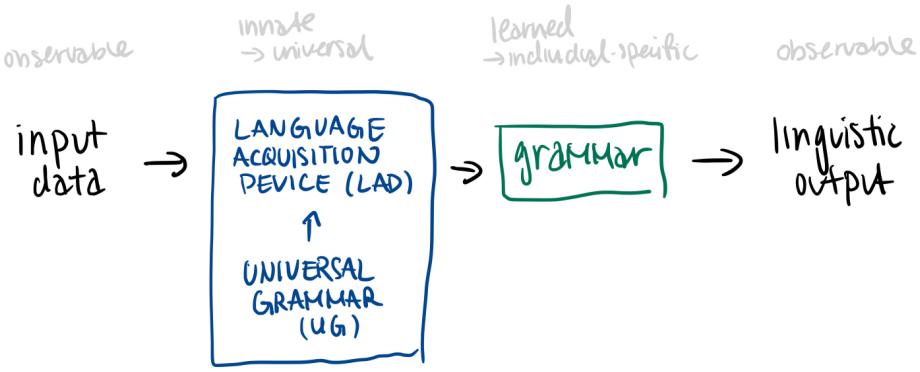


Figure 1: The system of language

At its core, the generative linguistics enterprise is concerned with this problem of learning. The explanatory adequacy of a theory is explicitly about whether that theory is learnable by a child on the basis of their input:

of event variable needed for CEN interpretations. This is the assumption that I do not adopt, primarily because nouns with eventive meanings pose no problem for learning, and since any architectural restrictions on grammar are really to serve the purpose of learnability by a child, I see no need to encode such a restriction. Part of this comes from the assumption of what Wood calls "Borer's Generalization": "a certain reading that nouns can have (the CEN reading) entails the existence of a morphologically related verb with that same meaning." (Wood 2023: 228) As discussed above, this generalization may not be true, and the fact that it holds in English may be understood for separate reasons.

> Although even descriptive adequacy on a large scale is by no means easy to approach, it is crucial for the productive development of linguistic theory that much higher goals than this be pursued. To facilitate the clear formulation of deeper questions, it is useful to consider the abstract problem of constructing an 'acquisition model' for language, that is, a theory of language learning or grammar construction… As a long-range task for general linguistics, we might set the problem of developing an account of this innate linguistic theory that provides the basis for language learning.
>
> (Chomsky 1965: 24–25)

Historically, the burden of explanation was placed on UG. The LAD was reduced to the seemingly trivial task of selecting one grammar, on the basis of input data, from the set of grammars fully specified by UG. Thus, a specific theoretical analysis of one speaker's grammar is taken to directly inform the shape of UG, and vice versa, with little further consideration to the process of language acquisition and the child's actual input.[39]

This was perhaps the only reasonable approach in the 1960s. The modern approach to linguistic analysis was still in its infancy, and very little was known about any language in these terms. There were little linguistic data available for analysis beyond one's own judgments and production. Naturally, even less was known about language acquisition, thus there was no grounds for a concrete theory of the LAD.

But the state of linguistics is different now. From the last six decades of work, we now know quite a lot about many languages of the world; along with this knowledge is the discovery that the diversity of languages cannot be reduced to limited parameters of variation. In addition, we now have an abundance of linguistic data that allows us to directly observe what a child's input looks like. Finally, we have an understanding of the process of language acquisition, including an emerging theory of the learning mechanisms that map a child's input data to learned linguistic knowledge.

In many ways, then, we have all the pieces to be able to put them together and directly answer the question of how language is learned. We can directly evaluate the learnability of a proposed grammar on the basis of our concrete theory of acquisition and the actual input data to a child. This can be done without the intermediary (and often vague) notions such as economy, or complicated stipulations on UG.

The present paper attempted to take such an approach. We directly examined a child's input data. We showed how our concrete learning mechanisms operate on these data to produce rules that comprise a speaker's grammar. We then showed how these rules generate the observable linguistic behavior of interest.

This approach is fully aligned with the model of the language system as in Figure 1, but we can make some further refinements. The important insight that enabled our resolution of the poverty of the stimulus problem in learning gerundive nominals was the dimension of time. The process in Figure 1 is usually assumed to be instantaneous: LAD+UG takes in input data, then returns a fully formed adult grammar.[40] In contrast, under the current approach, this process is iterative: it happens continually

[39] This is the program explicitly put forth by Chomsky (1965: ch. 1), following similar ideas in Chomsky 1957: ch. 6.

[40] It is perhaps more accurate to identify the difference between the present approach and other approaches to language acquisition not by the inclusion of time, but by the distinction between selection of a grammar and construction of a grammar. In most approaches (including both Principles & Parameters and Bayesian approaches), the problem of learning language is assumed to be the problem of selecting the correct grammar from an existing hypothesis space of possible grammars. Although some treat this learning as superficially a process, in that children may change their selection given new input, this process can really be collapsed into an instant of selection given all data. In contrast, the construction of a grammar is fully dependent on learning being a process over time: a child cannot learn gerundive nominals without having already learned ingP, nor can they learn ingP without having already learned VP. The more "basic" rules in language condition the acquisition of increasingly "complex" rules. This importance in the sequence of learning is what I mean by fully making use of the dimension of time. Treating language acquisition as construction, rather than selection, is also crucial to understanding

over the course of a child's development. At any given point, the output of learning are rules, which build on and add to a learner's existing repertoire of rules. This repertoire of rules is the grammar. A child's grammar can differ substantially from what we consider a fully formed adult grammar, depending on where they are in their development. One must then imagine the model in Figure 1 as taking place continually over time.

Furthermore, we have a refined understanding of the innate component, the LAD and UG. In our account of gerundive nominals, we made use of a concrete theory of learning and very light stipulations on UG. We made use of the general notions of Merge and syntactic categories: learning selectional rules are fundamentally about the learning what objects can be Merged, while syntactic categories are crucially what allows rules to apply to an abstraction over lexical items. These notions were not stipulated, but rather arose naturally from reasoning about the kinds of generalizations a child may make, in order to adequately capture the patterns in their input. In a way, this grounds the Minimalist goal to reduce UG to basic operations such as Merge in an actual theory of learning. As a first pass, the innate component may thus be understood as the mechanisms of learning and the human ability to process the world in certain ways (e.g. combinatorially).

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[Note: Appendices are under construction!]

A Learning mixed nominals

There is a third category of nominalization in English: mixed nominals like (51).

(51) John's reading of the book

In these cases, *-ing* is a true nominalizer and applies at the word level; it combines with a verb to form a noun. We can write the rule as in (52).

(52) **Rule:** *-ing* as a word-level nominalizer

$[_N$ -ing V] (productive over V)

To learn this rule, the child must encounter sufficiently many of the top 100 verbs occurring with *-ing* in the distribution of an N (e.g. in compounds, which children hear a lot of, and preceded by *the*). [Note: A more complete learning justification with counts will be added at a later date.]

The rule in (52) can generate mixed nominals, as shown by the derivation in (53).

(53) a. $[_N$ -ing read] (rule in (52))
b. $[_{NP}$ reading of the book] (prior knowledge of NPs)
c. $[_{DP}$ John's reading of the book] (prior knowledge of DPs)

Under this treatment, mixed nominals are identical to derived nominals, except the noun is productively formed by combining a verb and *-ing*. Indeed, mixed nominals behave like derived nominals. For example, mixed nominals display NP structure (e.g. modification by adjectives), and not vP structure (e.g. direct objects and subjects). This account of mixed nominals converges on existing approaches that attribute the differences between gerundive and mixed nominals to the attachment height of *-ing*, thus it covers the same empirical ground.[41]

B The perspective from diachrony

The interesting historical fact is that English developed gerundive nominals over the course of Middle English. This coincides with two developments of roughly that period. In EME, the gerundive ending was standardized to *-ing* while the previous alternative *-ung* died out. Also in EME, along with the loss of genitive case was the rise of the periphrastic genitive *of*, thus we get the standard modern mixed nominals. At the same time, the OE present participle *-ende* is gradually replaced by *-inge/-ing*, thus we see the merger of the gerundive and present participle suffixes. These developments suggest that we are on the right track by having both participle and gerunds as a crucial part of the evidence for learning gerundive nominals. However, the exact diachronic story still needs to be fleshed out. See Fanego 1996, 2004 for some relevant historical work.

[41] Under this treatment, the child really learns two rules involving *-ing*, one responsible for gerundive nominals (as well as other uses of ingPs with verbal structure) and the other responsible for mixed nominals (as well as other uses where V-*ing* is a noun, such as in compounds). The actual data, however, is somewhat messier than this partition suggests. The data consist of many examples that are ambiguous between the two (e.g. *your dancing*, *John's claiming that he saw Mary*, and *the rug cleaning company*). These ambiguous examples may thus be taken as evidence for both rules. Actual production also shows some overlap of the two rules; I found several examples of compounds with some verbal structure (e.g. *the going to bed book*) in my own data. So certainly more needs to be said to understand exactly what a child learns regarding *-ing* and its various uses.